MarriottConnect: An Integrated Student Information System with Decision Support Analytics for Marriott School

A Capstone Project Proposal
Presented to the Faculty of the
Information and Communications Technology Program
STI College Muñoz – EDSA

In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Technology

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March, 2026

ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH: MarriottConnect: An Integrated Student Information System with Decision Support Analytics for Marriott School

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In Partial Fulfilment of the Requirements
for the degree Bachelor of Science in Information Technology
has been examined and is recommended for Proposal Defense.

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APPROVAL SHEET

This capstone project proposal titled **MarriottConnect: An Integrated Student Information System with Decision Support Analytics for Marriott School**, prepared and submitted by **Jade Michael D. Godalle, Edson John R. Solitario, Francis Jay D. Raagas, Glezyl L. Gonzales,** and **Laurence Emmanuel M. Supangan**, in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, has been examined and is recommended for acceptance and approval.

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INTRODUCTION

Educational institutions increasingly rely on integrated information systems to maintain operational efficiency, data accuracy, and timely institutional decision-making. The effectiveness of how services are provided every day in schools where the operations are interconnected from an institutional administrative perspective, academic perspective and financial perspective is determined by the accuracy, reliability and availability of information. When there is only one time encoding of data through to its correct usage throughout the organization, the efficiency of workflow, reliability of reporting and stakeholders are able to receive their updates in a more timely manner. However, if school operations are still fragmented through various types of spreadsheets, paper logs and separate departmental records, then the school is at risk of having duplicate data, conflicting records and prolonged validation cycles.

The value of management information systems has long been associated with reducing procedural burden and improving reporting quality in multi-office environments. In contrast, fragmented systems often make institutions data-heavy but insight-poor, because data exists in large volume yet remains difficult to consolidate into usable, decision-ready outputs.

In the Philippine education context, the need for integrated school operations is even more pronounced because compliance-driven reporting and routine internal processing must be performed simultaneously. For Marriott School, this dual demand exposes the limits of disconnected tools and manual reconciliation practices. In response, this study proposes MarriottConnect as an integrated student information system with decision support analytics, designed to centralize operational records, reduce repetitive encoding, and provide leadership with structured evidence for planning and continuous improvement.

Project Context

Marriott School operates in a DepEd-governed environment where learner identity, enrollment reporting, and school-level operational records must remain synchronized through LIS-oriented procedures. According to Porlas et al. (2023), schools encounter higher validation burden and cross-office reporting inconsistencies when local MIS workflows are not tightly aligned with LIS structures. For Marriott School, this requirement is not merely technical compliance; it is an operational discipline that directly affects how registrar, finance, and academic offices coordinate daily decisions.

Appendix interviews confirm that this coordination remains difficult under fragmented tooling. The Academic Head explained that departments still maintain separate Excel files and frequently reconcile numbers manually before reports can be finalized. This condition creates a recurring pattern where staff attention shifts away from planning and student support toward repeated checking, correction, and consolidation tasks that should already be systematized.

The registrar workflow shows this problem in concrete terms. Mrs. Jocelyn Cleofe described enrollment and record updates as one-by-one processing with repeated cross-checking across spreadsheets and forms, especially during high-volume periods. According to Grepon et al. (2021), integrated school systems reduce delays and improve record reliability when admissions and records processing operate on one shared data structure. Kanona (2022) further emphasizes that centralized database-backed management improves retrieval speed and record integrity compared with manual, file-fragmented handling.

Finance operations demonstrate the same fragmentation in monetary form. Mrs. Corrine Avellanosa noted that payment updates often require manual cross-referencing before balances are confidently released, particularly when multiple transactions occur close together. According to Wulandari and Pinandito (2021), tuition-related decision support becomes more consistent when financial status logic is structured and system-driven rather than manually interpreted. Chai and Mostafa (2021) similarly report that integrated financial and student records reduce duplication and improve cross-user visibility.

Academic coordination is likewise affected. The registrar and teacher interviews indicate that schedule preparation, class-list alignment, and grade validation involve repetitive verification cycles before release. According to Muller et al. (2025), timetable quality and conflict reduction depend on explicit constraint handling rather than manual plotting. Zulueta et al. (2021) also show that process standardization and workflow discipline are critical in educational MIS environments, reinforcing the need for structured academic operations across sections, subjects, and grade submissions.

Stakeholder communication pressure further amplifies the issue. Parent-facing updates on grades, schedules, and billing are less effective when underlying records are delayed or inconsistently synchronized across offices. According to Navarra and Antonio (2025), web-based monitoring systems improve transparency and usability when authorized users receive timely, role-scoped visibility from one connected record environment. This aligns with Marriott School's need to reduce ad hoc follow-ups and provide dependable status communication to families.

At the leadership level, fragmented records also weaken decision support. Historical data may exist, but when data pipelines are disconnected, trend interpretation becomes late and less actionable. According to Esquivel and Esquivel (2021), enrollment-related decision support improves when institutional records are transformed into forecasting inputs rather than static archives. Hussain et al. (2023) further show that comparative forecasting approaches can provide stronger planning indicators for staffing, sectioning, and resource allocation.

For these reasons, the project context is best understood as an operational integration challenge: multiple offices depend on the same student lifecycle data but require different role-scoped actions. MarriottConnect responds by enforcing one validated data lifecycle from intake, posting, enrichment, and academic processing up to dashboard-ready analytics and governance outputs. In this architecture, data consistency, access control, and decision visibility are treated as interdependent design requirements.

Given this institutional context and evidence base, the following problem statements focus on the remaining strategic question: how an integrated, role-governed, and analytics-

capable system can continuously sustain enrollment consistency, finance synchronization, academic reliability, and leadership-ready decision support.

A deeper view of the present environment shows that data fragmentation is not limited to one office transaction. It propagates across the entire student lifecycle, from enrollment intake and payment posting to section assignment, attendance recording, and report generation. According to Zulueta et al. (2021), MIS environments in educational institutions fail when workflow standardization is weak, even when digital tools are available. This finding mirrors Marriott School's operational reality where records may exist in digital format but still require repeated reconciliation before they can be trusted for release, compliance, or planning.

In enrollment periods, the registrar must process a large volume of learner entries under time pressure while preserving LIS-oriented consistency. When intake records are encoded in disconnected trackers, every downstream update introduces a new opportunity for mismatch. According to Porlas et al. (2023), inconsistencies between LIS-facing structures and school-level MIS records create recurring validation burden across offices. For Marriott School, this means that what appears as a small intake encoding delay can later affect payment confirmation, class-list finalization, and administrative reporting confidence.

The record-continuity requirement extends beyond initial enrollment. Permanent records, promotion decisions, and student-departure handling all depend on the same validated learner identity baseline. If this baseline is fragmented, later academic and compliance outputs become harder to verify and slower to release. According to Kanona (2022), structured database-backed systems provide stronger retrieval reliability and control integrity than manual, file-fragmented record handling. This supports MarriottConnect's design principle that learner data should be encoded with continuity in mind rather than re-encoded at every stage of institutional processing.

Finance workflows add another layer of operational sensitivity because billing accuracy directly affects trust, cash-flow transparency, and release-dependent school transactions. The finance interview evidence indicates that balance confirmation often depends on

manual sequence checks when multiple payments occur in close succession. According to Wulandari and Pinandito (2021), tuition-related decision support improves when payment-status logic is formalized and system-driven. Chai and Mostafa (2021) also report that integrated student-finance data handling reduces duplicate effort while improving cross-role visibility, which is critical for parent communication and internal reconciliation timing.

Billing transparency is not only an accounting concern; it is a stakeholder-service concern. Families rely on timely balance updates, payment history visibility, and consistent communication channels, especially during high-demand periods. When billing records are delayed, schools face repeated follow-ups that consume office time and weaken confidence in published statements. According to Navarra and Antonio (2025), monitoring platforms improve communication outcomes when data is role-scoped, timely, and accessible through one connected environment. This aligns with MarriottConnect's integrated parent and finance views where institutional records and stakeholder-facing status updates remain synchronized.

Academic scheduling and class-list coordination reveal a similar structural challenge. Subject assignments, section constraints, and teacher availability must be reconciled in ways that prevent conflicts while preserving instructional continuity. Manual cross-checking can complete the task, but it often increases cycle time and revision overhead. According to Muller et al. (2025), timetable quality improves when constraints are explicitly modeled and resolved systematically, not manually improvised. In Marriott School's context, this supports the move toward structured schedule governance where academic planning outputs are reliable before dissemination to teachers, students, and parents.

Instructional workflows also require coherent links between attendance, grading, and verification stages. If class-level records are maintained in disconnected channels, teachers must repeatedly transfer and validate entries before submission, increasing the chance of latency and inconsistency. According to Zulueta et al. (2021), socio-technical alignment in MIS means process rules and system workflows must evolve together to sustain reliability.

MarriottConnect addresses this by treating attendance, score encoding, grade submission, and verification checkpoints as connected institutional processes rather than isolated encoding tasks.

Governance and access control requirements further shape the project context. Schools need role boundaries that prevent unauthorized edits while still allowing authorized users to access what they need on time. According to Kanona (2022), central database controls and role-scoped access are key to protecting information integrity in academic systems. The same governance logic is reflected in MarriottConnect through policy-driven access controls, audit visibility, and announcement/notification traceability, so transparency is achieved without sacrificing operational discipline.

Cross-role communication is another decisive factor in institutional performance. When offices and stakeholders rely on ad hoc messaging to confirm record status, coordination overhead grows and official channels lose authority. According to Chai and Mostafa (2021), integrated web-based school information environments improve communication consistency by keeping users connected to one authoritative dataset. This is why MarriottConnect pairs transaction modules with structured announcement and notification workflows: operational decisions and stakeholder communication should reference the same validated institutional data context.

At the leadership tier, the school's need is not only to store records but to interpret operational movement in time for planning. Enrollment growth, payment behavior, grade-submission cadence, and risk indicators all require timely synthesis to support proactive decisions. According to Esquivel and Esquivel (2021), decision-support value increases when institutional records are converted into forecasting-capable signals. Hussain et al. (2023) further demonstrates that trend modeling helps institutions prepare staffing and resource allocations more effectively than static historical review alone.

The implication for Marriott School is clear: reliable decision support depends on operational data continuity, and operational continuity depends on architecture that treats registrar, finance, academic, and stakeholder modules as one coordinated system. According to Grepon et al. (2021), integrated school information architectures improve

transaction flow and record reliability by reducing isolated processing channels. Combined with LIS-alignment evidence from Porlas et al. (2023), this reinforces that institutional consistency is produced through design discipline, not post-hoc manual correction.

Therefore, the project context must be interpreted as a systems-level coordination problem with compliance, governance, and stakeholder-service dimensions operating simultaneously. MarriottConnect is positioned to solve this by unifying module workflows, preserving role-based access discipline, and transforming validated operations into analytics-ready evidence for planning. In this framework, the problem statements below are not abstract design questions; they are direct responses to documented process friction and evidence-backed institutional needs observed in Marriott School and reinforced by recent literature.

Statement of the Problem

This study defines the operational problems of Marriott School based on interviews with key school stakeholders in the appendices, particularly the Academic Head (Appendix B.1), Registrar (Appendix B.2), Finance Officer (Appendix B.3), and Teacher (Appendix B.4). Their accounts consistently describe fragmented records, repeated reconciliation, delayed cross-office validation, and limited management visibility during high-volume cycles.

General Problem:

How can MarriottConnect be developed as an integrated student information system with decision support analytics for Marriott School?

According to the Academic Head interview (Appendix B.1), departments still maintain separate files and manually reconcile figures before reports can be finalized. According to the Registrar interview (Appendix B.2), enrollment updates are still processed one-by-one with repeated cross-checking across records. According to the Finance interview (Appendix B.3), payment posting and balance confirmation can be delayed when transactions occur in close succession.

Taken together, these interview findings show that Marriott School's general problem is workflow fragmentation across registrar, finance, and academic operations. Because offices validate the same learner information in different files and at different times, turnaround slows and management receives consolidated information later than needed for timely decisions.

Specific Problems:

- *How can a centralized registrar workflow be developed to capture enrollment intake, support SF1 upload, and enrich learner records by LRN to reduce duplicate encoding?*

  According to the Registrar interview (Appendix B.2), intake encoding, profile corrections, and enrichment checks are repeatedly validated across separate files before release. The same interview indicates that this verification burden becomes more severe during enrollment peaks, confirming duplicate encoding and repeated checking as persistent registrar-side bottlenecks.

- *How can integrated cashiering and billing workflows be developed to automate posting, update balances immediately, and keep ledgers and parent billing views synchronized?*

  According to the Finance interview (Appendix B.3), payment entries and balance confirmations often require manual cross-referencing before official status can be released. The interview further indicates that when multiple transactions are posted in close succession, reconciliation workload increases, which confirms cashiering and billing synchronization as a real institutional problem.

- *How can schedule and class-list workflows be developed to support academic planning while minimizing timetable conflicts and repetitive manual verification?*

  According to the Registrar and Teacher interviews (Appendices B.2 and B.4), schedule checks, class-list preparation, and grade-related validation are repetitive and labor-intensive. Their accounts show that academic coordination slows down

when subject, section, and teacher records must be manually verified across disconnected references before release.

- *How can secure student and parent portals be developed to provide authorized visibility to schedules, grades, and billing information?*

  According to stakeholder-facing interview evidence in the appendices, delayed record synchronization affects how quickly families receive complete updates. The Finance interview (Appendix B.3) indicates that parent balance communication is not always immediate, and related field accounts show that guardians often rely on follow-ups when records are still being reconciled across offices.

- *How can role dashboards and governance controls be developed to produce actionable indicators for enrollment trends, payment behavior, and operational risk monitoring?*

  According to the Academic Head interview (Appendix B.1), leadership reporting is slowed when department outputs come from separate files and require manual consolidation before review. This confirms that management-level indicators for enrollment movement, payment status, and academic monitoring are often delayed by fragmented data preparation practices.

Objectives

The objectives of this study are aligned directly with the general and specific problems above so that every development target has explicit operational and literature-based justification. Each objective is formulated as a direct development response to a corresponding problem condition validated by interviews and supported by recent RRL.

General Objective:

To develop MarriottConnect as an integrated student information system with decision support analytics for Marriott School.

This general objective is relevant because Marriott School requires a single operational

platform that can convert fragmented transactions into coordinated institutional outputs. MIS effectiveness is dependent on process alignment just as much as software capability according to Zulueta et al., (2021). Kanona (2022) has found that centralized data structures result in better consistency and therefore provide better control. Thus, the overall objective includes efficiency of everyday transactions, as well as decision-making at the leadership level.

Specific Objectives:

- *To develop a centralized registrar workflow that captures enrollment intake, supports SF1 upload, and enriches learner records by LRN to reduce duplicate encoding.*

  This objective directly responds to registrar-side duplication and validation delays observed in the appendices. According to Grepon et al. (2021), unified admissions and records pipelines reduce processing time and increase data reliability. Porlas et al. (2023) further supports LIS-aware alignment, making this objective essential for preserving data continuity from intake through reporting.

- *To develop integrated cashiering and billing workflows that automate posting, update balances immediately, and keep ledgers and parent billing views synchronized.*

  This objective is relevant because finance operations require immediate, trustworthy balance visibility to support both internal reconciliation and parent communication. According to Wulandari and Pinandito (2021), structured financial decision logic improves consistency in tuition management. Chai and Mostafa (2021) also supports integrated financial-record handling as a practical mechanism for reducing cross-check burden and improving transparency.

- *To develop schedule and class-list workflows that support academic planning while minimizing timetable conflicts and repetitive manual verification.*

  This objective addresses recurring scheduling friction in section and teacher

assignment workflows. Muller et al., (2025) report that when design constraints are identified and systematically addressed in a systematic manner, it leads to improvement of scheduling quality, with respect to conflict-free scheduling. Furthermore, Zulueta et al. (2021), note that schools need to create standardized procedures in order to achieve sustainable operations for Marriott in educational settings.

- *To develop secure student and parent portals that provide authorized visibility to schedules, grades, and billing information.*

  This objective is relevant because stakeholder trust depends on timely and role-appropriate access to validated school records. According to Navarra and Antonio (2025), monitoring platforms improve communication and usability when data visibility is immediate and structured. Chai and Mostafa (2021) similarly supports connected web access as a way to strengthen school-family information flow.

- *To develop role dashboards and governance controls that produce actionable indicators for enrollment trends, payment behavior, and operational risk* monitoring.

  This objective is relevant because leadership planning requires decision-ready indicators derived from validated operational data. Based on the research by Esquivel and Esquivel (2021), utilizing historical data in decision-making processes can enhance the value of decision-support systems. Hussain et al. (2023) further demonstrates how trend forecasting can aid in planning for resources and staffing, both of which contributes to a justification for developing KPI dashboards, governance controls, and analytics as primary functions within institutions.

Scope

The proposed system provides module-based access for seven core roles: super admin, admin, registrar, finance, teacher, student, and parent. Access is structured by institutional

responsibility and each module groups related actions into one operational unit.

The scope centralizes enrollment, finance, academic, governance, and stakeholder communication workflows while assigning each role to complete modules that include transaction handling, validation, and analytics interpretation.

The Registrar can access the following:

- **Enrollment Module -** Manage intake creation, profile updates, and payment-linked progression within one registrar workflow.
- **Student Directory Module -** Maintain searchable learner profiles, identity continuity, and authorized record retrieval.
- **SF1 Import Module** - Upload SF1 records and execute LRN-based enrichment aligned with LIS-oriented validation rules.
- **Permanent Records Module** - Maintain historical academic records and document continuity across school years.
- **Batch Promotion and Student Departure Module** - Process promotion outcomes and departure workflows under registrar controls.
- **Registrar Analytics** - Review intake movement, enrichment status, record-completion indicators, and registrar risk alerts.

The Finance Officer can access the following:

- **Cashiering Module** - Assess charges, accept payments, and post transactions with immediate balance reflection.
- **Student Ledger Module** - Track debit-credit movement, running balances, and learner-level financial history.
- **Transaction History Module** - Review transaction history and perform authorized void, refund, and reissue actions.
- **Billing Schedule and Due Allocation Module** - Manage due schedules and map posted payments to outstanding obligations.
- **Fee and Remedial Fee Configuration Module** - Maintain standard fee definitions and remedial subject fee policies.

- **Discount and Scholarship Rules Module** - Configure discount policies and apply student-level discount tagging controls.
- **Inventory Charges Module** - Manage finance-related item catalogs and charge integration in cashier workflows.
- **Financial Analytics** - Monitor collection trends, billing risk indicators, reminder outputs, and daily finance performance.

The Teachers can access the following:

- **Schedule Module** - Access assigned teaching loads, class timetables, and advisory schedule context.
- **Grading and Submission Module** - Configure rubrics, create assessments, encode scores, and submit quarter grades.
- **Attendance Monitoring Module** - Record SF2-aligned attendance statuses linked to subject-assignment context.
- **Advisory Conduct Module** - Encode conduct indicators and advisory observations for class-level monitoring.
- **Gradebook Validation Module** - Review grading completeness, submission status, and correction checkpoints.
- **Instructional Analytics** - Analyze class performance trends, completion alerts, and quarter progress indicators.

The Parents can access the following:

- **Schedule Module** - View child schedules, class timing, and section-related timetable context.
- **Billing Module** - Track dues schedules, payment history, and current balance snapshots.
- **Parent Analytics** - Review child-focused academic and billing indicators for timely family decisions.
- **Academic Records Module** - Access child's quarter and final grade records under role-scoped permissions.

The Students can access the following:

- **Student Academic Records Module** - Access own quarter and final grade records under role-scoped permissions.
- **Student Schedule Module** - View personal class schedules, subject timing, and advisory allocations.
- **Student Learning Analytics** - Review personal learning indicators, progress trends, and academic alerts.

The School Administrators (Academic Head/Principal) can access the following:

- **Academic Lifecycle Management Module** - Configure school-year dates, quarter transitions, and implementation controls.
- **Curriculum, Section, and Class List Module** - Manage subjects, section structures, adviser mapping, and roster outputs.
- **Schedule and Timetable Builder Module** - Build and adjust schedules with conflict-aware institutional controls.
- **Grade Verification and Compliance Module** - Enforce submission deadlines, review grade batches, and verify or return outputs.
- **Institutional Analytics** - Monitor enrollment, financial risk, academic status, compliance readiness, and KPI trends.

The Super Admin (IT Personnel) can access the following:

- **User and Access Administration Module** - Create accounts, assign roles, manage status, and enforce credential controls.
- **Permission Matrix Module** - Manage authorization views, role permissions, and governance policy controls.
- **Audit Logs and Announcement Module** - Review audit logs, publish announcements, and monitor recipient engagement.
- **System Configuration and Governance Analytics** - Control maintenance and portal settings, institutional preferences, and governance performance indicators.

Limitations

While MarriottConnect provides integrated coverage of registrar, finance, academic, and stakeholder workflows, the study remains bounded by institutional context, operational policy conditions, and data quality dependencies that influence measured outcomes.

For this study, the following limitations apply:

- System evaluation is grounded in one host institution (Marriott School); cross-school generalization requires additional deployments and comparative validation in other school environments.
- LIS-aligned processing depends on sustained institutional data discipline, including accurate intake encoding, timely SF1 uploads, and consistent LRN validation across operational cycles.
- Analytics outputs are based on available institutional historical records; forecasting and trend interpretation quality remains dependent on the completeness and consistency of encoded transactions.
- Operational performance remains dependent on reliable internet connectivity, available desktop access for guarded workflows, and sustained user adherence to defined role procedures.
- Policy, curriculum, and regulatory updates may require periodic configuration adjustments to preserve alignment between institutional operations and report-generation standards.

These limitations define the study context and evaluation boundaries while preserving the system's value as an integrated operational and decision-support platform.

Review of Related Literature/Studies/Systems

The present review of related literature and studies is designed to establish a strong theoretical and technical foundation for MarriottConnect by focusing only on sources that directly align with the school's documented operational problems. In this chapter, relevance is anchored on three validated issue clusters from the appendices: fragmented records across offices, delayed reconciliation in finance and registrar workflows, and limited decision-support visibility for planning.

To strengthen applicability, the proponents use ten studies published from 2021 onward, with five local Philippine references and five international references. Each source is evaluated not only on conceptual contribution, but also on direct fit to MarriottConnect modules such as enrollment intake and enrichment, cashiering and ledger workflows, schedule management, stakeholder portals, and analytics-enabled dashboards.

Local Studies

Designing and Implementing e-School Systems: An Information Systems Approach to School Management of a Community College in Northern Mindanao, Philippines

Grepon et al. (2021) conducted an applied research investigation into the creation of an integrated admission, record processing, and reporting system based on a unified digital work-flow. They qualified this research by demonstrating that if a school would work from a single set of shared data (rather than multiple sets of disconnected files) the time to process records and the reliability of records would improve. The study emphasized practical implementation behavior, including workflow continuity across offices and usability outcomes for non-technical institutional users.

This source is highly relevant to MarriottConnect because it directly supports the system's central design objective: convert fragmented office transactions into a single source of truth. In the appendices, the Registrar repeatedly described one-by-one updates and manual double-checking as a major bottleneck; Grepon et al. (2021) provides

empirical support for replacing that process with integrated registrar-to-finance-to-academic data continuity. The study also justifies the system's modular but connected architecture, where each office retains role boundaries while operating on shared records.

A Comparative Case Study on the Challenges Encountered by Philippine Private and Public Educational Institutions with their Existing Management Information System

In their study, Zulueta and colleagues (2021) analyzed challenges related to the implementation of Management Information Systems (MIS) that had been experienced in the Philippine Education System. The authors determined that there were ongoing issues relating to the standardization of processes, readiness of physical infrastructure, user adaptation, and operational support mechanisms based on real-world experiences. In addition to considering technical characteristics of MIS applications as a parameter for measuring performance, the authors also evaluated the effectiveness of these systems through a socio-technical lens (interaction of the organizational work structure and the quality of MIS system design) and emphasized that coordination of these two elements was vital for effective implementation.

For MarriottConnect, this study is relevant because the appendices show that current delays are not caused by a single office failure but by cross-office coordination friction. Zulueta et al. (2021) supports the capstone's staged rollout logic and reinforces the need for clear role-specific workflows, consistent operational rules, and adoption-oriented implementation practices. It also strengthens the rationale for including governance-related controls and maintainability planning, not just transactional modules.

A Machine Learning Based DSS in Predicting Undergraduate Freshmen Enrolment in a Philippine University

Esquivel and Esquivel (2021) demonstrated how machine-learning-based decision support can transform historical enrollment records into forecasting outputs usable for institutional planning. The study showed that data-driven prediction methods can reduce reliance on manual trend estimation and improve preparation for upcoming academic

periods. Its contribution is particularly valuable because it connects routine data storage with forward-looking administrative action.

This is directly applicable to MarriottConnect's analytics direction, where enrollment trend interpretation is expected to support section planning, staffing preparation, and capacity decisions. In the appendices, school leadership noted difficulty producing consolidated trend insights from dispersed records; Esquivel and Esquivel (2021) provides a methodological bridge from transactional history to planning intelligence. The study therefore supports the inclusion of forecasting-ready analytics in the administrative dashboard model.

Comparative Analysis on the Efficiency of LIS Vis-a-Vis SMIS: A Case Study

Porlas et al. (2023) compared the efficiency of DepEd LIS processes and school-level MIS workflows, emphasizing the operational consequences of inconsistent synchronization between compliance-oriented and institution-level records. The study highlighted that when local school systems do not align cleanly with reporting structures, institutions face additional validation burden and inconsistent data interpretation across offices.

This finding is highly relevant to MarriottConnect because the system is intentionally designed for LIS-aware processing rather than isolated internal encoding. The staged model used in the capstone intake, cashier posting, and SF1/LRN-based enrichment directly addresses the synchronization challenge discussed by Porlas et al. (2023). The source strengthens the justification for treating enrollment data quality and reporting alignment as core architectural requirements, not post-processing tasks.

Efficient Student Monitoring and Data Tracking System

The research of Navarra & Antonio (2025) showed that by using a web-based approach for tracking and monitoring students, teachers, students and parents obtained positive results in terms of both functionality and usability. The findings of the study demonstrate the importance of having timely access to information about the school, having clear lines

of communication between institutions and parents, and being able to access timely information about their child's academic progress.

For MarriottConnect, this is strongly relevant to stakeholder-facing modules. Appendix responses indicate that parents often experience delayed or incomplete updates, while offices rely on fragmented communication channels to resolve record issues. Navarra and Antonio (2025) supports the system's parent and student portal strategy, where authorized users can access updated academic and billing-related information through one secure platform rather than informal or delayed communication paths.

International Studies

Designing Decision Support System for Midwifery Students' Tuition Fees Problem

Wulandari and Pinandito (2021) proposed a decision support model for tuition delinquency management using structured decision logic. Their study moved beyond simple balance viewing by showing how financial status classification can be translated into actionable policy options for school management. The research demonstrated that decision-support logic can improve consistency and speed in handling financially sensitive student cases.

This aligns directly with MarriottConnect's finance goals, particularly delinquency identification and actionable payment-status interpretation. In appendix interviews, the Finance office described delays caused by manual cross-referencing of ledgers during multiple payment events. Wulandari and Pinandito (2021) supports the inclusion of structured delinquency analytics so finance decisions are guided by standardized rules and timely data, not fragmented manual reconciliation.

Bright Kids Tuition Centre Management Information System

Mostafa and Chai (2021) created a web-based information management system, which provided an interconnected environment in which many different parties (students, parents, staff) could obtain access to the same set of records. The results of their study showed that using a common platform for records that were being shared led to some

benefit, such as reduced duplication of data entries; and by providing better means of communicating, it enabled a more effective means of accessing and managing those records by the different types of users.

This is relevant to MarriottConnect because the capstone addresses the same fragmentation pattern observed at Marriott School: separate operational trackers, repeated validation, and delayed reporting. Chai and Mostafa (2021) supports the system's web-first architecture and validates the design choice of linking registrar, academic, and stakeholder views to one connected data model. It also reinforces the need for clear role-scoped access in shared operational systems.

Student Information Management System for Baghdad College of Economic Sciences University (SIMS-BC): A Case Study

Kanona (2022) examined the transition from manual student record handling to a secure database-driven system, emphasizing gains in retrieval speed, consistency, and data protection. The study showed that procedural efficiency and information integrity improve when record operations are backed by structured database controls rather than ad hoc file management.

For MarriottConnect, this source supports both architecture and governance components. The appendix also states that offices spend an inordinate amount of time currently validating records in a variety of formats and files; Kanona (2022) proposes a method for handling records by utilizing a central database with role-based controls to alleviate that burden and meet the overall security posture of the school where only authorized users have access to sensitive student and finance information while protecting that information from unauthorized modifications.

Student Gross Enrolment Ratio Forecasting: A Comparative Study Using Statistical Method and Machine Learning

Hussain et al. (2023) compared forecasting approaches such as ARIMA and LSTM for educational enrollment trends, demonstrating that historical institutional data can be transformed into more reliable forward-looking indicators. The study contributes

methodological depth by showing how comparative modeling can guide strategic interpretation instead of relying on one static projection style.

This is directly relevant to MarriottConnect's decision-support layer, where enrollment movement is intended to inform staffing, section planning, and resource preparation. In the appendices, leadership concerns repeatedly pointed to difficulties in trend interpretation because records were dispersed and manually consolidated. Hussain et al. (2023) supports the capstone objective of converting validated operational history into structured planning insight through dashboard-ready analytics.

Real-world University Course Timetabling at the International Timetabling Competition 2019

Muller, Rudova, and Mullerova (2025) analyzed real-world timetabling constraints and evaluated solver behavior under institution-scale scheduling conditions. The study highlighted how timetable quality depends on explicit constraint modeling, reproducible optimization logic, and measurable conflict handling rather than manual timetable plotting.

This is highly relevant to MarriottConnect's schedule-related direction. Appendix findings describe scheduling checks as labor-intensive and prone to delay when performed through manual verification. Muller et al. (2025) provides strong technical support for moving toward conflict-aware scheduling logic, where resource constraints are handled systematically and schedule outputs become more consistent across teacher, student, and parent views.

Synthesis

Across the ten selected studies, a common conclusion emerges: schools achieve better operational reliability when data is centralized, workflows are role-defined, and reporting is generated from validated transactions rather than from fragmented records. The local

literature confirms contextual fit for Philippine school operations, including LIS-aligned processes, implementation barriers, and stakeholder communication realities.

International literature complements this by supplying deeper methodological support for tuition decision support, secure information management, enrollment forecasting, and constraint-driven scheduling. Together, the local and international sets provide both contextual grounding and technical depth for MarriottConnect's major modules.

Most importantly, the literature aligns with evidence from Marriott School's own appendix interviews. Registrar one-by-one encoding concerns, finance reconciliation delays, and leadership reporting difficulties all map directly to the problem domains addressed by the selected sources. For this reason, the RRL foundation not only supports conceptual design but also validates the system's implementation priorities and module-level direction for the capstone.

TECHNICAL BACKGROUND

Overview of Current Technologies Used in the System

MarriottConnect is implemented as a modern web-based school information system that centralizes institutional records, enforces role-based operations, and delivers analytics-ready outputs for decision support.

The backend uses Laravel 12 on PHP 8.5 with PostgreSQL as the primary relational database. Authentication and access security are handled through Laravel Fortify, while role middleware and route-group policies control authorized module access.

The frontend uses Inertia.js v2 with React 19 and TypeScript to support structured, responsive workflows per role. Tailwind CSS v4 and Shadcn component patterns are used to maintain consistent interfaces across registrar, finance, academic, and governance pages.

For analytics and reporting, shared chart wrappers support line, bar, area, and pie visualizations, enabling trend interpretation through standardized dashboard payloads.

Prototyping Model

The study follows an iterative prototyping model to ensure that system behavior continuously reflects validated operational requirements from school stakeholders.

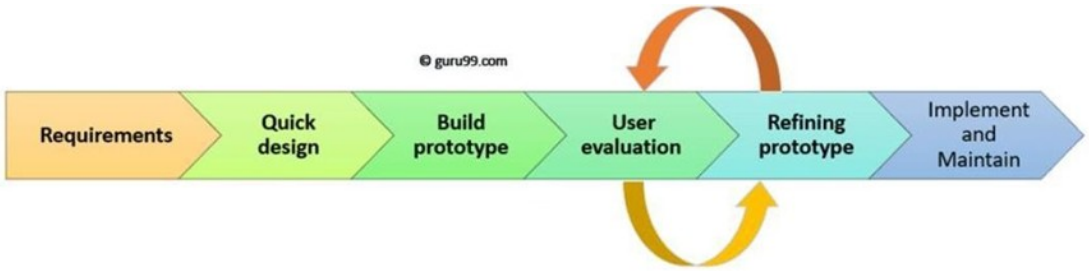

Figure 1. Prototype Model

Reference model: iterative prototyping cycle for requirement validation, controlled implementation, and continuous refinement.

The model applies the following stages to maintain user-centered and operationally aligned development:

1. 1. Requirements Gathering and Analysis

   Interviews, process mapping, and role-task validation are used to identify bottlenecks in enrollment, cashiering, scheduling, grading, and reporting workflows.

2. 2. Quick Design

   Initial page flows, data mappings, and module interaction logic are prepared based on verified responsibilities and system boundaries.

3. 3. Build Prototype

   A working prototype is produced for core operational modules, including registrar intake, finance posting, instructional grading, and role dashboards.

4. 4. Initial User Evaluation

   Key users from registrar, finance, academic, and administration evaluate usability, output validity, and workflow fit against actual office practice.

5. 5. Refine Prototype

   Evaluation feedback is applied iteratively to improve data accuracy, process speed, and interface clarity until acceptance criteria are satisfied.

6. 6. Implement and Maintain

   Validated modules are rolled out for controlled use, followed by user orientation, policy alignment, and maintenance planning for sustained operations.

Calendar of Activities

August - Initial data gathering, stakeholder interviews, and baseline workflow

documentation were completed to establish the project problem context.

September - Scope definition, chapter drafting, and requirements consolidation were aligned with observed institutional workflows.

October - Module architecture and functional design were revised based on consolidated interview insights and process-validation results.

November - Final validation inputs were integrated, and technical and functional sections were updated to reflect the complete implementation design.

Gantt's Chart of Activity

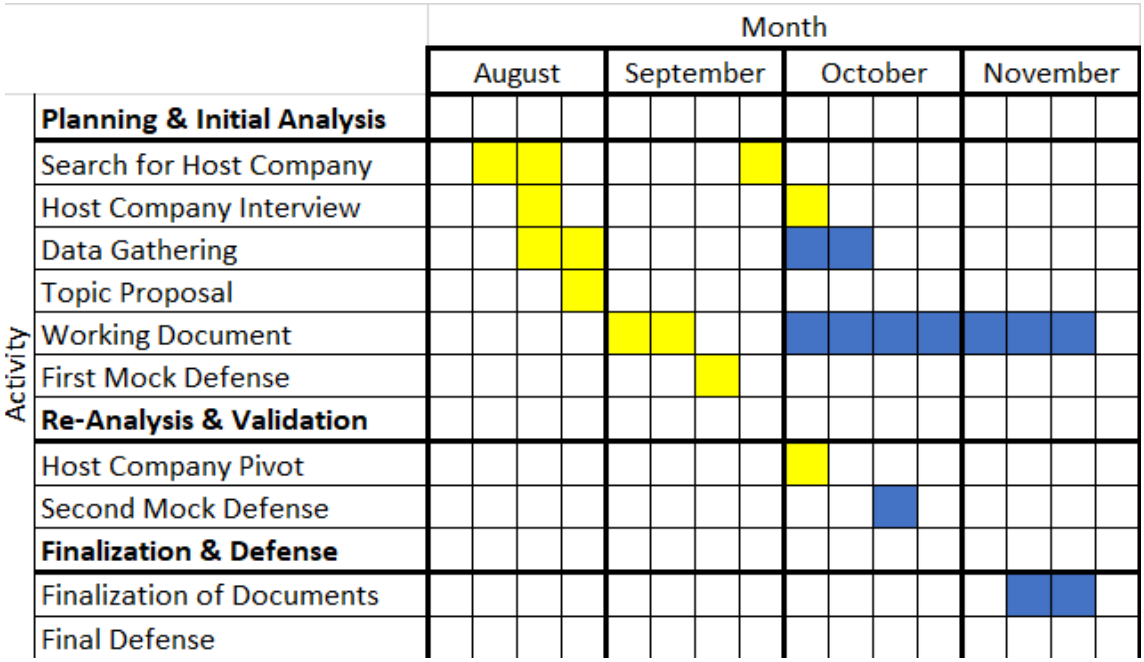

Figure 2: Gantt's Chart of Activity

Resources

Hardware Requirements

The following resources are required to host, access, and operate MarriottConnect across school offices and stakeholder portals.

- Application/Database Server: Dedicated or virtual infrastructure with sufficient CPU, memory, and storage for concurrent transactions and analytics workloads.
- Client Devices (Mobile): Smartphones or tablets for parent and student access to authorized schedule, grade, billing, and notification views.
- Client Devices (Web): Desktop or laptop devices with updated browsers for registrar, finance, admin, and teacher operations.

Software Requirements

- Server Runtime: PHP 8.5 and Laravel 12 with secure environment configuration for web requests, queues, and scheduled tasks.
- Database Management: PostgreSQL as the relational data engine for centralized and consistent institutional records.
- Backend Services: Laravel Fortify for authentication, role middleware for access control, and audit-safe processing for governance actions.
- Frontend Stack: Inertia.js v2, React 19, and TypeScript for structured SPA-like role workflows and maintainable component architecture.
- Interface and Visualization Framework: Tailwind CSS v4, Shadcn component patterns, and chart wrappers for consistent UI and analytics presentation.
- Development Environment: Composer, Node.js tooling, Vite, and Visual Studio

Code for implementation, integration, and deployment preparation.

- Quality and Collaboration Tools: Figma for interface planning, with Pest, PHPUnit, Pint, ESLint, and Prettier for quality assurance and code consistency.

Requirements Analysis

This section defines the requirements needed to address Marriott School's validated operational constraints through MarriottConnect, with emphasis on centralized data handling, role-based process control, and analytics-backed decision visibility.

A. Requirement

Marriott School requires one integrated platform that captures enrollment intake, supports finance posting, enables grading workflows, and provides role-scoped dashboards while maintaining LIS-aligned learner record continuity.

B. Business / User Requirement

The system must minimize repetitive encoding and manual reconciliation by ensuring that validated records flow consistently across registrar, finance, teacher, student, parent, and governance modules under controlled permissions.

System Integration for Student Information and Operations

1. The system must allow registrar intake data to become reusable input for finance and academic workflows.

2. The system must support staged enrichment through SF1 upload and learner matching by LRN.

3. The system must synchronize balances, schedules, and grade visibility across authorized role portals.

4. The system must provide audit-aware and role-scoped access to protect sensitive transactions and records.

Web-Based Administrative and Analytics Workspace

1. Authorized administrators must manage academic controls, sections, schedules, and class visibility through one secure interface.

2. Administrative users must review KPI cards, alerts, and trends that summarize real-time operational conditions.

3. Administrative outputs must support school-year planning, staffing decisions, and workflow correction.

C. System Requirements

Major System Capabilities

1. The system operates as a role-based web platform accessible through modern desktop and supported mobile browsers.

2. The system centralizes learner, enrollment, grading, and finance records under one validated data model.

3. The system enforces role-based access for super admin, admin, registrar, finance, teacher, student, and parent users.

4. The system supports registrar intake, student directory maintenance, and SF1 upload processing.

5. The system supports finance cashiering, ledger updates, due allocation, fee configuration, and transaction history actions.

6. The system supports teacher grading workflows including rubric setup, activity encoding, score entry, and quarter submission.

7. The system provides dashboard analytics through KPI cards, alerts, and trend visualizations.

8. The system preserves governance through settings control, audit logging, permissions visibility, and announcement management.

Major System Conditions

1. Each user must authenticate using valid credentials tied to an authorized role.

2. Stable internet connectivity must be available for synchronized updates and dashboard refresh cycles.

3. Registrar intake and finance posting must complete before downstream enrichment and reporting workflows finalize.

D. System User Characteristics

Super Admin and Admin users manage governance, configuration, curriculum context, compliance controls, and planning policies.

Registrar users handle learner intake, enrollment queue operations, SF1 enrichment, and permanent-record continuity.

Finance users process cashier transactions, maintain ledgers, manage assessments, and monitor collection-related indicators.

Teacher users manage instructional schedules, grading tasks, attendance monitoring, and class-level performance checks.

Student users access personal academic records, schedules, and learning indicators under restricted permissions.

Parent users access child-related schedules, grades, and billing information through secure portal controls.

E. Functional Requirements

1. Manage Student Records

1.1 The Registrar shall create and maintain learner profiles and enrollment queue records.

1.2 The system shall support SF1-driven enrichment and LRN-based learner matching logic.

2. Manage Cashiering and Assessment

2.1 Finance users shall post cashier transactions and maintain learner-level ledger entries.

2.2 The system shall update balances immediately after valid posting events.

2.3 The system shall support fee, discount, and inventory-linked assessment definitions.

3. Manage Grading and Submission

3.1 Teachers shall configure rubrics and encode graded activities by section, subject, and quarter.

3.2 The system shall compute weighted outputs and retain draft and final submission states.

3.3 The system shall support authorized review flow for quarter completion and submission status.

4. Manage Schedule Visibility and Coordination

4.1 Authorized users shall access role-specific class schedules generated from maintained academic assignments.

4.2 Schedule views shall remain consistent across teacher, student, and parent role contexts.

5. Manage Scheduling and Class List Preparation

5.1 Admin users shall configure planning inputs through sections, subjects, and faculty assignment context.

5.2 The system shall support conflict-aware schedule preparation and class-list validation workflows.

6. Provide Decision Support Analytics

6.1 The system shall process operational metrics for enrollment movement, payment behavior, and risk monitoring.

6.2 The system shall visualize trends using standardized chart-ready payloads for management decisions.

7. Generate Operational Reports

7.1 The system shall provide structured outputs for enrollment, finance, and academic summaries.

7.2 Reports shall remain consistent with authorized data sources and institutional formatting policies.

F. Non-Functional Requirements

Operational Requirements

1. The system must remain accessible through supported web clients for office and stakeholder use.

2. The system requires stable connectivity for synchronized transactions and dashboard updates.

Performance Requirements

1. Posting and grade-related updates must reflect promptly after validated submissions.

2. The platform should maintain reliable responsiveness during normal school-hour usage.

Security Requirements

1. Credentials must be protected through secure authentication handling and password hashing.

2. Authorization rules must prevent cross-role access to restricted modules and records.

3. Governance-relevant actions must be traceable through audit-ready logging mechanisms.

Usability Requirements

1. Interfaces must present clear task flows aligned with each role's routine operational responsibilities.

2. Pages must maintain readability and consistency across desktop and supported mobile contexts.

Maintainability Requirements

1. The system must support controlled updates to school-year settings, fee structures, and role configurations.

2. Core modules must remain maintainable so policy updates and institutional enhancements can be integrated without destabilizing existing workflows.

Requirements Documentation

MarriottConnect is structured as an integrated module ecosystem where registrar, finance, academic, governance, and stakeholder services operate on one validated institutional dataset.

The Registry and Enrollment module captures intake entries, maintains learner profiles, and supports SF1 upload with LRN-based reconciliation for LIS-aligned continuity.

The Cashiering and Finance module manages assessment logic, payment posting, ledger updates, transaction actions, and billing visibility outputs for authorized users.

The Academic Planning module supports school-year controls, curriculum mapping, section management, schedule preparation, and class-list coordination.

The Teacher Operations module provides rubric configuration, graded activity encoding, attendance tracking, score submission, and gradebook validation workflows.

The Student module provides secure personal access to schedules, grades, and learning indicators based on authorized academic records.

The Parent module provides child-focused visibility to schedule, academic performance, billing status, and school-issued notifications.

The analytics layer standardizes KPI cards, alerts, trend cards, and action links so each role receives decision-ready summaries from validated transactions.

Trend rendering supports line, bar, area, and pie charts through normalized payload contracts to preserve readability and interpretation consistency.

Governance services include user management, permission controls, audit logging, announcements, and system setting enforcement for operational integrity.

These integrated requirements position MarriottConnect as a centralized, role-governed, and decision-support-oriented platform for Marriott School operations.

Design of Software, System, Product, and/or Processes

The software design of MarriottConnect follows a role-centered, modular architecture that separates registrar, finance, academic, stakeholder, and governance concerns while maintaining one shared institutional data model.

At the backend layer, Laravel route groups and middleware enforce role boundaries so each user enters only authorized modules and actions. This design prevents cross-role data misuse while preserving workflow continuity across connected processes.

At the frontend layer, Inertia.js with React and TypeScript provides component-based pages that map directly to role workflows. Shared UI patterns are used to keep task flow predictable for high-frequency operations such as intake, posting, grading, and verification.

The data design is relational and continuity-oriented: learner identity, enrollment state, financial transactions, schedules, attendance, and grades are linked through validated references so updates are reusable across modules without repeated encoding.

Process design is built around staged operational flow. Registrar intake initializes learner context, finance posting updates balance state, enrichment validates identity continuity, and academic modules consume authorized records for schedule and grading outputs.

Analytics design uses standardized KPI, alert, and trend payload contracts so dashboards across roles can render consistent indicators while still reflecting role-specific context. This allows school leadership and operational staff to interpret metrics from one coherent data pipeline.

Governance design integrates authentication, permission visibility, audit traceability, and configurable system controls. Together, these controls support accountability, policy alignment, and secure institutional operation.

Overall, the product design treats MarriottConnect not as isolated feature pages but as one coordinated school operations platform where transaction quality and decision visibility are engineered as connected outcomes.

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APPENDICES

APPENDIX A. RESOURCE PERSONS

MR. ALEXANDER F AVELLANOSA

ACADEMIC HEAD

MARRIOTT SCHOOL

MRS. JOCELYN M. CLEOFE

REGISTRAR'S OFFICE

MARRIOTT SCHOOL

MRS. CORRINE P. AVELLANOSA

FINANCE OFFICE

MARRIOTT SCHOOL

MRS. FE MERCEDES M. CAVITT

TEACHER

MARRIOTT SCHOOL

MR. DAVEY

GUARD

MARRIOTT SCHOOL

APPENDIX B. TRANSCRIPT OF INTERVIEWS

B.1 Academic Head

Interviewer: Jade Michael D. Godalle
Interviewee: Mr. Alexander F. Avellanosa

Interviewer: Good afternoon po, Sir Alex. I'm Jade po ulit. Nandito po ulit kami para sa interview. Sir, diba yung previous natin, gumagamit po kayo ng Excel or manual files. Bakit po sir, hanggang ngayon gumagamit pa rin po kayo ng Excel?

Interviewee: Actually, i-clarify ko lang. Excel na nasa G-Drive po 'yan. So, manual siya, pero nasa cloud, naka-share. Alam mo yun, parang okay na rin siya sa simula, kasi lahat makakakita, makaka-access, pero mahirap i-manage. Kasi ang dami-dami ng files, bawat department may kanya-kanyang Excel, at minsan kailangan i-compare manually. Kahit nasa cloud, kung sabay-sabay nagta-trabaho, hindi mo na talaga ma-track kung sino nag-edit, paano nag-change yung data. Talagang hassle. Kaya nga kahit may Excel, hindi pa rin siya ganap na efficient.

Interviewer: Sir, yung mga nag-offer sa inyo, nag-offer ng ano? Alam po ba nila yung problem kaya nag-offer sila ng ganong system sa inyo?

Interviewee: Ah, yun. Dami nag-offer sa amin ng enrollment systems or iba pang system. Pero honestly, di namin sigurado kung fully nila naintindihan yung ganyang specific problems namin. Kasi ang hirap talaga, iba-iba yung situation namin. Yung mga ino-offer nila, parang standard package lang sa ibang schools. Hindi naman sila talaga fit sa workflow namin. Kaya kung gagamitin namin yun, baka mas magulo pa. Talagang kailangan namin maintindihan muna bago i-adopt.

Interviewer: Sir, ano po ang mahirap na naranasan kapag sabay-sabay na nag-edit o nag-access ng parehong file ang iba't iba?

Interviewee: Oo, tama. Mahirap talaga. Halimbawa, sabay nag-e-edit si Finance at si Registrar sa parehong Excel file sa Google Drive. Pag may mali, kailangan i-trace kung sino nag-edit, ano yung nagbago, saan nagkamali. Parang, ini-examine

mo isa-isa yung cells. Ang hirap talaga kasi minsan nawala yung data o may nag-change na hindi mo alam. Kailangan balik-balikan, i-check ang history ng bawat entry. Nakakapagod talaga sa oras at effort.

Interviewer: Yes, sir.

Interviewee: Kaya nga minsan, kahit may Excel, parang manual pa rin kasi kailangan pa rin ng constant monitoring, checking, at coordination. Hindi siya automatic na nag-a-update o nagma-manage ng conflicts.

Interviewer: Sir, gaano kadalas nagkakaroon ng error or inconsistency sa student records?

Interviewee: Student records? Dito, medyo old school talaga. Talagang handwritten ang initial input, tapos i-encode pa sa Excel. Hindi naman madalas mag-error technically, kasi careful kami sa pag-check, pero ang problema, sobrang tagal i-verify. Ang workflow, kailangan mo i-double check, i-compare ang entries, minsan printed copy, minsan digital copy. Nakakapagod. Kapag may sabay-sabay na nag-update, may mga points na parang nawawala o hindi aligned agad.

Interviewer: So, sir, sabi niyo po old school yung process niyo. May nangyayari po ba na duplicate ng student records?

Interviewee: Ah, yung present students? Hindi naman technically duplicate, pero minsan may confusion sa data integrity. Halimbawa, minsan hindi ka sigurado kung ang grade na na-encode na ba or hindi pa. Parang, "San ko na ba nilagay yun?" o kaya "Na-print ko ba ito?" Kaya kailangan ulit i-check manually. Nakakapagod talaga.

Interviewer: Sir, paano naapektuhan ng errors and delays na ito ang operational supervision ng school sa mga students?

Interviewee: Talagang sobra ang oras na binibigay namin para maiwasan ang delay. Halimbawa, pag may problem sa grade or enrollment, nire-recall namin ang student or teacher para i-verify. Kung hindi ganito, aabutin ng delay ang students sa card release or sa enrollment confirmation. Talagang nakakapagod ang manual management.

Interviewer: Like, paano po yun sir ngayon? Di ba sir, kuhaan po ng card ngayon, then di nakuha ng students, bukas po pwede silang bumalik?

Interviewee: Next week na yun. Kasi may rules na dapat parent ang kukuha ng card sa first quarter. Kung wala yung parent, hindi makukuha ng student. Kaya minsan may delay talaga. Kahit ganoon, kailangan ng effort para masigurong updated at verified lahat ng records bago ibigay sa students.

Interviewer: Sir, sa experience niyo, madalas ba kayong makaranas ng delays or errors sa communication between registrar, teachers, and finance?

Interviewee: Oo, sobrang common. Kasi lahat manual. Minsan, nagbayad na yung student, pero hindi pa na-update sa Excel ng registrar. Minsan, nagpalit ng section ang student, late na na-inform ang teacher. Kaya kailangan pa rin ng extra coordination at checking.

Interviewer: So, sir, paano niyo hinahandle yung ganung instances kapag may miscommunication?

Interviewee: Usually, pinapatawag namin yung involved departments. Halimbawa, may mismatch sa payment o grades, pinapa-verify, tapos pinapapirmahan para documented. Pero minsan, ang effort talaga malaki kasi need mo pa rin manually i-check lahat ng files at records.

Interviewer: Sir, ano ang mga challenges kapag kinokolekta ang attendance mula sa iba't ibang klase o section?

Interviewee: Ah, isa sa pinaka-problema namin ay manual talaga. May attendance si teacher, may attendance ang class president, tapos kailangan i-compare. Kaya minsan, hindi pareho ang record. Halimbawa, isang araw may na-miss ang teacher mag-record, tapos may na-record yung student officer. Pag kinokolekta mo ulit, kailangan i-check isa-isa, i-verify kung sino ang tama. Nakakapagod at prone sa delay.

Interviewer: Sa anong katagal bago makonsolidate ang attendance data para sa reports?

Interviewee: Usually, weekly. Lahat ng teachers at officers magkocollect mula Monday to Friday. Pagkatapos, nire-review at kino-consolidate para sa monthly report. Nakakapanood ka na lang minsan, tapos ayun, i-check mo ulit kung may kulang or mali. Talagang time-consuming.

Interviewer: Sir, may mga pagkakataon bang kulang o mali ang attendance records?

Interviewee: Oo, kadalasan dahil sa rushed yung teachers. Minsan nagtuturo pa sila, tapos nakakalimutan mag-check ng attendance. Kaya kailangan i-recall the next day or follow-up manually. Kung minsan, hindi na maalala kung sino talaga ang present nung araw na yun. Kahit may manual record ang student, kailangan pa rin i-double check.

Interviewer: Laiway po ba yung parents para mag-update ng attendance ng anak nila?

Interviewee: Meron. Pero mostly, nakikipag-communicate kami sa parents kapag may problema. Halimbawa, absent ng dalawang araw, tinatawagan namin para malaman kung okay lang ba yung student o may kailangan ba. Talagang close ang relationship namin sa parents at students, para informed sila. Hindi namin basta pinapalampas.

Interviewer: Ah, so tatawagan niyo po yung parents?

Interviewee: Oo, usually kinabukasan kung may absence. Pero karamihan ng parents, sila rin ang nagre-report agad kung absent ang anak. Dapat nga, close ang communication para maiwasan ang gap sa information.

Interviewer: Sir, paano ina-update ng teacher staff yung mga errors sa attendance?

Interviewee: Usually, tinatawagan muna yung involved teacher or student officer. Kung kayang i-adjust agad, ginagawa namin bago pa ma-finalize ang report. Talagang manual process pero kailangan matapos bago ma-print yung final attendance card. Kung hindi, baka hindi aligned yung data, tapos magkakaroon ng confusion sa student or parent.

Interviewer: Sir, paano nabe-verify ng school kung present o absent ang student sa mismong araw?

Interviewee: Meron silang manual record. Teacher records, student records, tapos may Blackboard updates na nakalagay kung sino ang absent or late. Halos lahat ng info need i-verify sa dalawang sources. Nakakapagod, kasi kung may missing entry, kailangan i-follow up.

Interviewer: Sir, ano ang epekto kapag na-delay o mali ang attendance?

Interviewee: Mostly, nagkakaroon ng concern ang parents, nagfe-feedback kung hindi accurate. Pero, since mabilis naming na-verify at na-update, rarely maabot sa parents ang error. Pero effort talaga, marami kaming kailangan gawin para maiwasan yung discrepancy.

Interviewer: Sir, sa palagay niyo ba makakatulong ang RFID-based attendance system para mapabuti ang process nito?

Interviewee: Oo, definitely. Kasi automatic yung logging. Halimbawa, pumasok na yung student, maitatala agad sa record. Parents automatically makakatanggap ng notification. Mas mabilis, mas efficient. Pero kahit maliit lang yung loophole, halimbawa nagpasok pero hindi talaga pumapasok sa class, iikot pa rin kami para ma-monitor. Pero overall, malaking tulong sa speed at accuracy ng tracking.

Interviewer: Sir, ano ang mahirap kapag kailangan kontakin ang mga magulang lalo na sa emergency o absences?

Interviewee: Pag nahihirapan kaming kontakin, kailangan dumaan sa phone, landline, o messenger. Minsan, kailangan puntahan personally sa bahay. Pero most of the time, okay lang kasi close ang relationship namin sa parents. Malalapit lang naman sila, kaya reachable.

Interviewer: Sir, paano naapektohan ng delay sa communication ang tiwala ng mga magulang sa school?

Interviewee: Mostly okay lang. Medyo flexible ang parents kasi alam nilang gagawin namin ang effort para ma-update sila. May mga ways kami na reachable sa kanila: e-mail, cellphone, Facebook Messenger, kahit Facebook page namin. So kahit may delay, aware sila at nagtitiwala sa communication process namin.

Interviewer: So sir, sa palagay nyo po, bakit makakatulong ang automatic notification para sa mga clients?

Interviewee: Oo, malaking tulong. Automatic eh, diba? Immediate na yung info sa parents kung pumasok o hindi ang anak. Mas iwas human delay o human error. Halimbawa, RFID tap lang, automatic na naitatala at natutext yung parent.

Interviewer: Sir, di naman po kayo gumagamit ng spreadsheet o manual computation para sa tuition po, no sir?

Interviewee: Hindi, hindi naman po. Ang ginagamit namin, Excel pa rin pero naka-Google Drive. So may automatic na computation doon, pero mostly still manual pa rin sa pag-check.

Interviewer: Sir, paano po ninyo titiyak na accurate at updated ang tuition records?

Interviewee: Ah, dito kami very careful. Marami nagche-check. May cashier, may finance head, at ako rin minsan tumitingin para ma-monitor. Yung mga resibo, naka-save sa Google Drive, so kahit may errors, mabilis ma-verify. Kailangan lang talaga constant monitoring.

Interviewer: Sir, paano naapektohan ang manual na computation ng workload ng accounting office?

Interviewee: Siyempre, mas matagal. Kasi one by one yung check, tapos may manual adjustments pa. Mas marami ang trabaho, mas matagal bago matapos. Kung automated, mas mabilis, pero since manual, kailangan talagang tutukan ng staff.

Interviewer: Sir, bakit minsan hirap ang mga magulang o estudyante na makita ang kanilang payment status?

Interviewee: Siguro dahil nasa iba't ibang sources pa yung data. May naka-Excel, may naka-email, tapos may hard copy. Kaya minsan, kailangan nila mag-follow up para makumpirma. Kung centralized man, mabilis na makita, pero ngayon, medyo fragmented pa yung info.

Interviewer: Sir, bakit patuloy pa rin gumagamit ng magkakaibang tools sa bawat department?

Interviewee: Kasi, hindi naman lahat nakasama sa same workflow. Halimbawa, registrar may Excel, finance may iba, at teachers may sariling records. Kaya, bawat department may sariling tool, kasi need nila immediate access sa sariling data. Hindi kasi one-size-fits-all yung workflow nila.

Interviewer: Bakit sa tingin nyo, sir, ay risky ang pag-store ng student data sa shared folders?

Interviewee: Oo, risky kasi maraming nakakakita, maraming nakaka-access. Kahit na may permissions, minsan may editing conflicts, tapos may chance na ma-misplace or ma-overwrite yung data. Kailangan laging ma-monitor at i-double check.

Interviewer: May mga pagkakataon bang nawala o nangbura ang files?

Interviewee: Meron, pero bihira. Usually dahil may nag-edit na hindi napansin o na-misplace lang. Kapag ganun, babalik ulit sa process, i-verify ulit, at i-reprint kung necessary. Kaya nakakapagod talaga minsan, kasi isa-isa kailangan i-check.

Interviewer: Paano nyo pinipigilan ang unauthorized access sa files?

Interviewee: Sa Google Drive, nakaset kung sino lang ang pwedeng mag-view o mag-edit. Tinitiyak naming limited lang yung access, at sino ang editor, sino ang viewer. So may kontrol pero kailangan palaging bantayan.

Interviewer: Bakit mahalaga magkaroon ng secure password-protected system para dito?

Interviewee: Para ma-minimize yung risk na may makaka-access ng data na hindi dapat. Importante na alam namin sino lang ang may right mag-edit o mag-view ng file.

Interviewer: Paano naapektohan ng manual process ang productivity ng teachers at staff?

Interviewee: Medyo mabagal talaga, kasi one by one ang process. Mas maraming steps, mas matagal matapos. Pero sa kabilang banda, mas mabusisi. Nakikita mo lahat ng details. Pero kung mas mabilis at organized, mas maraming nagagawa sa parehong oras.

Interviewer: Gaano kayo nakakaranas ng delay sa paggawa ng reports dahil sa disorganized data?

Interviewee: Hindi naman totally disorganized kasi may sariling process kami. Pero syempre, kapag manual, mas matagal ang reporting. Kailangan i-verify lahat ng data bago i-finalize. Kaya minsan, may delays kahit planado ang workflow.

Interviewer: Paano naapektohan ng kakulangan ng automation ang accuracy at timeliness ng data?

Interviewee: Dito nakakaapekto sa speed at accuracy. Kadalasan, mas matagal mag-update, mas prone sa delay, kasi manual pa rin. Kailangan i-double check, i-

verify, tapos maayos bago ma-release. Kung automated, mas mabilis at mas ma-track.

Interviewer: Sir, sa inyong palagay, ano ang pinakamalaking challenge sa kasalukuyang sistema ng school?

Interviewee: Marami. Pero ang pinakamalaking problema, talaga, yung kahirapan sa coordination ng iba't ibang departments. Halimbawa, kung enrollment, finance, at teachers, bawat isa may sariling paraan, tapos need i-merge manually. Minsan, nagka-conflict yung numbers student count, tuition payment, attendance. Kailangan mag-check isa-isa para ma-align.

Interviewer: Paano nito naapektohan ang trabaho ninyo at ang students?

Interviewee: Mas maraming oras ang nauubos sa pag-verify ng data kaysa sa actual na tasks. Teachers at staff, kailangan mag-double check ng entries, kaya delay sa reporting. Students, minsan natatambakan ng late updates sa grades, attendance, at payments. Ang stress, hindi sa work lang, pati sa operations ng school.

Interviewer: Sir, bakit sa tingin nyo nanatili pa rin ang mga problema na ito hanggang ngayon?

Interviewee: Siguro kasi complex ang system namin. Maraming tools at manual processes. Lahat connected sa ibang tasks. Wala pang streamlined workflow na madaling ma-access ng lahat. Kaya hanggang ngayon, ganito pa rin.

Interviewer: Ano ang tingin nyo na solusyon para maayos ang mga problemang ito?

Interviewee: Siguro, kailangan ng mas malinaw at standardized workflow. Halimbawa, kung paano kino-collect at chine-check ang attendance, paano nagha-

handle ng tuition, paano nagva-verify ng student records. Kasi kapag consistent ang process, mas mabilis maayos ang error, mas mabilis ma-access ang info, at mas maiiwasan ang conflicts sa data. Mas mapapabilis ang trabaho at maiiwasan ang hassle sa lahat.

B.2 Registrar's Office

Interviewer: Jade Michael D. Godalle
Interviewee: Mrs. Jocelyn M. Cleofe

Interviewer: Paano niyo po mini-maintain at ina-update ng records ng mga students ninyo?

Interviewee: Sa ngayon, manual pa rin kami sa Google Sheets at Excel. Kailangan namin i-update isa-isa ang bawat record, tsaka i-double check rin para siguradong tama ang lahat ng details. Kapag maraming students, tumatagal ang proseso kasi may sections na nagkaka-overlap, kailangan i-verify ang bawat entry at minsan kailangang i-crosscheck sa ibang forms.

Interviewer: Madalas ba kayong makaranas ng errors o data inconsistency sa records ng students?

Interviewee: Oo, minsan may inconsistency sa data lalo na kapag maraming updates sa parehong oras. Kadalasan, kailangan i-compare ang entries sa Excel at Google Sheets, tapos siguraduhin na parehong information ang nakalagay sa lahat ng files. Hindi ito agad naayos, kaya minsan tumatagal bago maging final.
Interviewer: Included po ba dito yung duplicated entries, late submissions, o miscommunications, ma'am?

Interviewee: Oo, minsan may overlapping entries, tapos minsan late rin ang submission ng requirements. Halimbawa, yung enrollment forms o proof of residency, minsan dumarating lang after deadline. Kaya minsan delayed din ang pag-update ng records, tapos kailangan i-follow up sa parents at teachers.

Interviewer: Pero ma'am, may instance na po ba na nag-duplicate yung record?

Interviewee: May pagkakataon, lalo na kapag maraming information ang naipapasa sa parehong oras at sa parehong files. Kailangan talaga naming i-verify bawat entry para siguradong hindi mag-overlap. Minsan tumatagal ang verification kasi hindi lahat ng data ay available sa iisang lugar.

Interviewer: Gaano katagal bago niyo ma-verify o ma-update ang record ng isang estudyante kapag may concern?

Interviewee: Depende sa dami ng students at dami ng information na kailangang i-check. Kung maraming forms at kailangan i-verify sa ibang files, minsan umaabot ng ilang araw bago ma-finalize ang record. Kailangan din i-review ang bawat section para siguradong consistent ang lahat.

Interviewer: Paano po kayo nakikipag-coordinate sa accounting office at mga guro tungkol sa student data?

Interviewee: Kadalasan, manual ang coordination. Kapag may issue, tatawag kami sa teacher o staff, minsan kailangan i-email o i-meet para klaruhin ang information. Kailangan pang i-follow up kung may kulang o inconsistent, at minsan kailangan ulitin ang process hanggang sa kumpleto ang data.

Interviewer: Madalas ba magkaroon ng delay sa pagkuha o pagpapasa ng data like grades, tuition info, enrollment list, etc.?

Interviewee: Oo, minsan may delay kasi naghihintay kami sa kumpletong forms o approvals mula sa iba't ibang departments. Kadalasan, kailangan i-double check bago ma-finalize, kaya minsan natatagal ang proseso ng isa o dalawang linggo depende sa dami ng students.

Interviewer: Ano yung pinakamahirap na process ng coordination sa ibang department?

Interviewee: Minsan mahirap i-track kung updated na ba ang data ng bawat department, lalo na kapag may pinapasa-pasang forms o schedules. Kailangan i-verify sa bawat step bago ma-finalize ang record. Halimbawa, pag nag-e-encode ng grades, kailangan siguraduhin na kumpleto ang forms at na-review ng teachers bago ma-final.

Interviewer: Paano po yun ma'am? Kunyari sa teachers po, parang may head po sila, tapos bago yung papasa sa registrar office, parang ganun po ba?

Interviewee: Oo, may hierarchy pa rin. Pero kahit na na-review na ng principal, kailangan pa rin namin i-check ang bawat entry sa registrar office bago ma-finalize. Minsan, kailangan i-crosscheck ang forms sa manual files, kaya tumatagal ang proseso lalo kapag maraming students.

Interviewer: Paano niyo dinidistribute ang class schedules?

Interviewee: Ngayon, manual pa rin. Kailangan i-check ang schedule ng bawat teacher para siguradong walang conflict sa oras at subjects. Kung may conflict, kailangan i-adjust isa-isa, tapos i-inform lahat ng teachers at principals. Kapag maraming subjects at sections, challenging talaga ang process at tumatagal ang finalization.

Interviewer: Manual po ba yun?

Interviewee: Oo, mahirap at time-consuming. Kailangan i-review bawat schedule, i-compare sa bawat teacher at section, tapos siguraduhin na walang overlap. Kahit maliit na pagbabago, kailangan i-adjust ang buong schedule para consistent sa lahat.

Interviewer: May mga chance po ba na nalilito o nadedelay ang mga studyante o guro dahil sa schedule updates?

Interviewee: Oo, minsan nagkakalituhan lalo kapag may pagbabago sa schedule. Kailangan i-update at ipaalam sa lahat ng teachers at students, tapos i-double check para siguradong consistent sa bawat section. Kapag hindi na-update agad, nagkakaroon ng confusion at delay sa klase.

Interviewer: Gaano katagal kayong gumagawa ng reports tulad ng enrollment, summer grades, report, or class list?

Interviewee: Sa grades, karaniwan 3-4 days para ma-encode lahat, tapos may additional time pa para i-review at i-verify ang bawat entry. Kapag maraming students, minsan mas matagal kasi kailangan siguraduhin na consistent at kumpleto ang data bago i-release.

Interviewer: Yung pwede na pong i-distribute?

Interviewee: Mga 3-4 weeks bago ma-release kasi kailangan pang i-review ng lahat ng teachers at principal ang final data. Lahat ng tests, quizzes, at exams ay i-encode muna, tapos tinitingnan ulit bago gawing final.

Interviewer: Sa enrollment summary?

Interviewee: Weekly namin kino-check at manu-manong nirerecord kung sino ang nag-enroll at nagpa-reserve sa isang linggo. Kailangan i-update bawat list para siguradong accurate, at minsan tumatagal lalo kapag maraming students ang nag-enroll sabay-sabay.
Interviewer: Sa class list naman?

Interviewee: Dinidiretso manually. Kapag nag-enroll na ang student, saka lang nilalagay sa section. Kailangan siguraduhin na tama ang lahat ng information at consistent sa records, kaya minsan tumatagal lalo kapag maraming students.

Interviewer: Balik po tayo sa grade reports sa releasing of grades, manual po ba yan o online?

Interviewee: Manual pa rin. Pupunta ang parents dito para kunin ang card. Kailangan i-check muna ang lahat ng files bago ibigay para siguradong updated at complete ang record ng student.

Interviewer: Wala po kayong soft copy?

Interviewee: Meron during pandemic sa Google Classroom, pero ngayon, hard copy talaga ang ginagamit. Kailangan i-prepare isa-isa at siguraduhing consistent sa lahat ng records.

Interviewer: So yung parents lang po ang may access sa grades?

Interviewee: Oo, parents lang. Guardian pwede rin kung mas matanda. Hindi binibigyan ang students ng access para i-maintain ang consistency at privacy ng records.

Interviewer: Ano po ang pinaka-challenge sa sistema ng record management sa registrar office?

Interviewee: Pinakamahirap ay ang manual na pag-track at pag-update ng lahat ng student information. Kailangan i-check sa maraming files bago maging final, lalo na kapag maraming students at sections. Lahat ng entries ay mano-manong nirereview para siguradong consistent at kumpleto.

Interviewer: How about sa pag-distribute?

Interviewee: Hindi mahirap sa konting students, pero kapag marami, challenging talaga. Kailangan i-prepare at i-review isa-isa para siguradong consistent sa lahat ng sections at teachers.

Interviewer: Sa pag-e-encode, na-aaccess po ba ng ibang teachers yung files o kayo lang?

Interviewee: Office lang ang may access. Dalawa lang kami, at kailangan i-review lahat ng steps bago ma-finalize ang record. Kailangan i-check bawat entry para siguradong consistent sa lahat ng files.

Interviewer: Hindi po kayo sabay?

Interviewee: Hindi. Isa-isa kami nag-e-encode at nag-aayos, tapos saka ipapasa para sa printing. Kailangan maayos at kumpleto bago ma-finalize, kaya tumatagal talaga.

Interviewer: Yun lang po Ma'am. Thank you po!

Interviewee: Thank you rin!

B.3 Finance Office

Interviewer: Jade Michael D. Godalle
Interviewee: Mrs. Corrine P. Avellanosa

Interviewer: Paano niyo po nirerecord ang mga tuition at balances sa mga students?

Interviewee: Sa ngayon, manual pa rin kami gamit ang Excel. Lahat ng student balances at payments, nakalagay sa spreadsheet, tapos ini-store namin sa Google Drive para centralized sa amin. Dito nakikita ang lahat ng student data enrollment,

payment mode, at balance. Kapag may nag-enroll, ini-encode namin ang details sa sheet, tapos automatic nagkakalculate yung running balance at total enrollment. Sa parehong sheet, nakikita rin namin ang total payment. Kasi connected sa Drive, kahit sino sa finance team na may access, makikita rin agad. Pero kasi manual pa rin lahat, kailangan i-verify at i-update isa-isa, kaya minsan tumatagal lalo kapag maraming students.

Interviewer: Ah, okay po. Parang centralized na siya sa registrar, cashier, at financier?

Interviewee: Oo, tatlo kami ang directly nagma-manage nito, tapos meron din kaming external accountant. So apat ang nakaka-access sa Drive. Pero kahit centralized, manually pa rin i-update at i-check bawat record para siguradong consistent ang lahat.

Interviewer: Tatlo po?

Interviewee: Oo. Dalawa sa amin nagre-record, tapos ang accountant nagdo-double check ng summary. Pero lahat ng data manually pa rin pinapasa at i-verify.

Interviewer: Then, ma'am, madalas po ba kayong makaranas ng errors o delay sa pag-update ng payment record?

Interviewee: Oo, may delay minsan kasi manual ang proseso. Kapag may online payment or on-site payment, kailangan i-update isa-isa sa sheet. Kapag maraming payments sabay-sabay, may delay bago makita sa system ang updated balance ng student. Kadalasan, may time gap sa pag-record ng transaction at sa availability ng data sa spreadsheet.

Interviewer: Ay ma'am, paano po ma-determine na hindi na-update?

Interviewee: Usually, nakikita namin sa statement of account. May mga different modes of payment GCash, BDO, on-site. Pag hindi pa na-update ang payment sa sheet, hindi pa rin reflected sa statement ng student. Minsan tumatagal bago marecord, depende sa processing at pag-update sa Google Drive. Kaya may time lag sa availability ng complete data.

Interviewer: May mga errors po. Ma'am, ano po yung karaniwang dahilan ng discrepancy o pagkakaiba ng records?

Interviewee: Ang common issue talaga ay timing ng pag-update. Kapag may multiple payments sa parehong araw, minsan hindi agad na-update sa sheet ang lahat. Kailangan pa i-crosscheck sa sequence ng transactions. Isa pa, may mga manual checks sa series ng ORs at transactions para siguradong kumpleto ang data. So minsan delayed ang reflection sa summary hanggang ma-verify.

Interviewer: Then ma'am, gano'ng katagal ma-update ang balance ng estudyante pagkatapos mabayad?

Interviewee: Yung on-site payments, usually agad na-update sa same day kasi dito mismo sa office binabayad. Sa online transactions naman, depende sa processing ng bank, usually maximum three days bago fully ma-update sa spreadsheet at statement. Kailangan rin i-verify ang details bago ma-finalize para consistent sa lahat ng reports.

Interviewer: Ay ma'am, paano po yan? Kunyari gusto ng parents makita ulit yung binayaran nila at yung balance nila, papunta pa ba sila dito?

Interviewee: Pwede naman. May monthly statement of account na ibinibigay sa kanila. Nakikita nila doon lahat ng payments at balances. Kapag may discrepancy,

puwede silang mag-email o tumawag para ma-clarify agad. Pero kailangan pa rin i-verify sa records namin bago ma-update ang statement.

Interviewer: May POS po ba kayo, ma'am?

Interviewee: Wala, Excel-based talaga lahat. Walang point-of-sale system ngayon, kaya manual pa rin ang recording at updating ng payments.

Interviewer: So, ma'am, parang anytime, makikita ng parents ang status nila?

Interviewee: Hindi agad. May monthly schedule kami ng statement updates. Kaya kadalasan, nag-aantay muna sila ng monthly statement para makita ang current balance at payments.

Interviewer: Gano'ng kadalas nagtatanong ang mga magulang at estudyante tungkol sa tuition balance o payment status?

Interviewee: Mostly monthly, kapag may statement of account. May mga parents rin na mas proactive, nagche-check ng payment online, pero karamihan naghihintay muna ng official statement para makita ang full details.

Interviewer: Ano ang karaniwang paraan ng pagbibigay nyo ng updates, text, printed statements, or in-person?

Interviewee: May messaging system kami para automatic makapag-notify kapag may statement. Pero may printed statement din na ibinibigay monthly. Lalo na sa mga parents na mas comfortable sa printed copy, parang mas madaling ma-verify nila. May iba rin pumupunta para personal check.
Interviewer: I-email lang po talaga? Wala po kayong messenger?

Interviewee: May email at messaging, pero mas madalas na-print out. Kasi dati,

email lang, minsan hindi nababasa ng parents, lalo na yung lola't lolo. Kaya mas maayos na may physical copy.

Interviewer: Ma'am, may mga pagkakataon ang updates kaya nagkakaroon ng misunderstanding sa payment?

Interviewee: Oo, minsan may confusion. Pag may transactions na hindi pa fully na-update sa sheet, may parents na nagtatanong kung na-record na. Kailangan naming i-verify ang data sa spreadsheet bago ma-finalize.

Interviewer: Anong klaseng errors po yung nae-encounter niyo, ma'am?

Interviewee: Yung mga delay sa pag-record ng payments at hindi agad na-update sa statement. Pag automatic at real-time, agad lalabas ang updated balance at transaction history.

Interviewer: Paano niyo po ginagawa mga financial reports tulad ng payment summaries, outstandings, balances?

Interviewee: Excel-based din. May Statement of Account kami, balance sheets, at summary per section at grade level. Ina-update daily ang payments, tapos may outstanding balances na nakalista. Binibigyan rin ng copy ang teachers para makatulong sa follow-up. May external accountant din na nag-summarize ng total financial statement, pero manual pa rin ang proseso.

Interviewer: Gano'n po katagal yan, ma'am? Estimated lang po.

Interviewee: Daily kami nag-uupdate. May weekly summary, tapos monthly reports. So araw-araw may review at update, tapos weekly at monthly check sa external accountant. Pero bawat step, manual pa rin, kaya time-consuming.

Interviewer: Ma'am, paano po kayo nakipag-coordinate sa registrar o admin kapag may updates sa enrollment or payment data?

Interviewee: Google Drive ang ginagamit namin. Pwede agad makita ng registrar ang total balances. Lalo na kapag exam at may card release, kailangan ma-check kung may outstanding payments bago ma-release. Kung hindi updated, puwedeng ma-delay yung card release.

Interviewer: May tanong po pala ako sa grades, pag release po ba ng card, required na bayad na yung estudyante?

Interviewee: Oo, required na bayad bago ma-release ang card. Ito ang pang-hold ng school.

Interviewer: May mga chance bang nagkakaroon ng problema sa synchronization ng data sa ibang department?

Interviewee: Wala naman major, kasi centralized sa akin lahat ng payments. Pero minsan, kapag may delay sa update, puwedeng magkaroon ng problema sa registrar, halimbawa, hindi updated ang balance sa card release.

Interviewer: Then ma'am, last question po. Ano ang mga challenges na gusto niyong mawala kung magkakaroon ng automated financial system?

Interviewee: Madami. Gusto ko maiwasan ang delays sa pag-update, magkaroon ng transparency sa parents, at ma-monitor agad ang payments. Lalo na kapag marami na kaming students 500 plus na ngayon. Kasama rin yung books, uniforms, supplies. Iba't ibang payment modes. Kung automated, mabilis na makaka-generate ng reports, maayos ang tracking ng enrollment count at payments, at mas efficient ang daily work ko.

Interviewer: Like, ma'am, iisa-isa yun yung hahanapin pa?

Interviewee: Oo, isa-isa. Kaya sobrang busy kapag enrollment. Kapag automated, immediate na lalabas lahat ng details, ORs, history ng payments, previous year info, delinquent or not. Mas madali at mas efficient.

Interviewer: Okay na po, ma'am. Thank you.

B.4 Teacher

Interviewer: Jade Michael D. Godalle
Interviewee: Mrs. Fe Mercedes M. Cavitt

Interviewer: Paano ninyo kasalukuyang nirerecord ang attendance ng mga estudyante?

Interviewee: Ah, ngayon po, manual pa rin talaga. Gumagamit kami ng attendance sheets na printed. Kada pasok sa classroom, dala namin yung sheet tapos isa-isa naming tine-tick kung sino ang present. Medyo matrabaho lalo na sa unang period kasi kailangan talagang i-check. Minsan inaabot pa bago makumpleto, lalo na kapag may mga late.
Interviewer: Ah okay, manual po?

Interviewee: Oo, manual talaga. At dahil papel, kailangan pang ilipat-lipat minsan nasa classroom, minsan naiipon sa adviser's table bago ma-process.

Interviewer: May times po ba na nagkakaroon ng errors or delays sa pag-submit ng attendance?

Interviewee: Nagkakaroon ng delay kapag maraming klase sa isang araw. Yung attendance sheet kasi kailangan pang i-collect at i-review, lalo na kung may excused o may follow-up. Dahil hiwa-hiwalay yung sheets per section, medyo tumatagal

bago ma-compile. Hindi naman malaki ang delay, pero mabagal yung flow dahil bawat update nililipat pa sa ibang record.

Interviewer: Ma'am, ano ang mga dahilan kung bakit minsan naging mahirap i-manage ang attendance logs?

Interviewee: Mahirap siya kapag marami kang hawak na klase. Kahit familiar ka na sa students, kailangan mo pa ring isa-isahin lalo na pag may absent o bagong transfer. Tapos dahil papel lahat, kailangan mo pang i-store nang maayos. Kapag may hinahanap na past attendance, babalikan mo pa yung lumang sheets, kaya natatagalan talaga.

Interviewer: Ma'am, gaano po kahalaga sa inyo na real-time makita ng school at ng magulang ang attendance ng bata?

Interviewee: Importante po, lalo na para sa parents. May mga bata kasi na pumapasok pero hindi agad nalalaman ng magulang kung anong oras sila dumating. At may mga students din na minsan lumiliban. Kung real-time, mas mabilis yung coordination, at mas nakakatulong siya sa safety ng bata.

Interviewer: Ma'am, paano niyo po karaniwang sinasubmit ang grades sa registrar?

Interviewee: May deadlines po kami. Pero bago ma-submit, dadaan pa sa encoding, checking, at computation. Yung grades kasi nakahiwalay written works, performance tasks, exams kaya kailangan pang pagsama-samahin. Kapag maraming students, tumatagal talaga bago maging final yung records.

Interviewer: Gumagamit po ba kayo ng tools tulad ng Excel para isubmit sa registrar?

Interviewee: Oo, gumagamit kami ng Excel. Pero hindi lahat agad na-eencode kasi hindi naman kami laging nasa computer. Madalas sinusulat muna sa papel habang nasa classroom, tapos lumilipat na lang sa Excel kapag nasa faculty na. Kaya parang doble trabaho pa rin.

Interviewer: So ma'am, wala pong case na na-delay yung paper submission sa registrar?

Interviewee: Meron pa rin minsan, lalo na kapag may revisions o may kulang pa sa requirements. Hindi naman sobrang late pero naaapektuhan yung bilis ng finalization kasi kailangan munang kompletuhin lahat bago isumite.

Interviewer: Paano po kung late na talaga ang submission?

Interviewee: Kapag late, mas maraming naaapektuhan kasi naka-depende yung reports ng admin sa grades namin. Sa deliberation pa lang, matagal na ang checking. Pag marami yung students, mas matagal pa bago matapos kaya may mga pagkakataon na halos dikit sa deadline yung submission.

Interviewer: Ma'am, nahihirapan ba kayo mag-update ng student records kapag nagkakaroon ng revisions?

Interviewee: Oo, lalo na kung revisions galing sa FAPE o admin. Maraming kailangan i-update names, sections, requirements. Dahil hiwa-hiwalay yung copies sa adviser, subject teachers, at registrar, isa-isang ina-adjust para pare-pareho. Kapag may kulang pang dokumento, balik na naman sa manual checking.

Interviewer: Paano po kayo karaniwang nakikipag-ugnayan sa mga magulang tungkol sa performance ng estudyante?

Interviewee: Mostly sa PTA or conferences. Pero kung may urgent, tinatawagan namin. Ang challenge lang, minsan matagal bago maiparating yung information dahil sa schedules ng parents at teachers.

Interviewer: Ano sa palagay nyo ang pinaka-challenges sa current system ng record keeping?

Interviewee: Sa ngayon, yung dami ng papeles at records talaga ang mabigat. Iba-iba ang documents attendance, grades, requirements, behavior notes. Pag may kailangan ang admin o parent, hahanapin mo pa kung nasaan yung specific file. Minsan nasa classroom, minsan nasa faculty. At kapag marami kang sections, mas matagal yung paghanap at pag-ayos ng data. Hindi naman dahil may maling ginawa, pero dahil sa dami ng hawak na records, mabigat talaga sa oras at proseso.

B.5 Security Guard

Interviewer: Jade Michael D. Godalle

Interviewee: Mr. Davey

Interviewer: Paano niyo nalalaman kung sino na ang mga estudyante sa school, lalo na sa umaga, at kung sino ang wala pa?

Interviewee: Nakikita ko lang sila sa gate habang pumapasok. Isa-isa ko tinitingnan kung sino ang andun na at sino pa yung wala. Minsan, kapag marami, medyo mahirap i-track lahat agad kasi sabay-sabay din pumapasok yung iba.

Interviewer: May record ba kayo kung anong oras pumasok ang isang estudyante, o naka-base lang kayo sa obserbasyon?

Interviewee: Observation lang talaga. Wala kaming exact record ng oras, kaya minsan nakadepende lang sa mata ko kung sino ang dumating. Kung gusto ko i-note, kailangan ko pang isulat manually sa notebook, pero kapag busy sa gate, kadalasan hindi ko na agad nasusulat.

Interviewer: Kapag dumating ang mga service gaya ng tricycle o school van, paano niyo tinitiyak na tama ang mga batang sinasakay nila, lalo na kapag sabay-sabay?

Interviewee: Tatanungin ko muna yung driver at iche-check kung tama yung pangalan ng bata sa listahan. Kapag sabay-sabay ang mga service, medyo nagkakagulo, kaya pinapila ko sila at tinatawag isa-isa yung bata para siguradong tama.

Interviewer: Kapag may bagong driver o pinalitang service, paano niyo chine-check kung authorized silang kumuha ng bata?

Interviewee: Tinitignan ko muna sa list sa admin kung nakalista sila. Kung wala, hindi ko pinapayagang sumakay yung bata. Kailangan talaga ma-verify para maiwasan yung problema sa parents o sa school.

Interviewer: Paano niyo nalalaman kung nakauwi na o na-pick up na ang bata, at paano niyo pinapaalam sa admin o teacher?

Interviewee: Tinitingnan ko kung sumakay na yung bata sa tricycle o van. Kapag marami sabay-sabay, minsan medyo nakakalito, kaya kailangan ko i-check isa-isa. Pagkatapos, sinasabi ko sa admin o teacher kung sino na ang nakalabas o nasundo, minsan nililista rin sa logbook para may reference.

Interviewer: May mga pagkakataon ba na nag-aalala ang magulang kung nakauwi na ang bata, at madali ba silang ma-update?

Interviewee: Oo, madalas tumatawag o nagte-text sila para siguraduhin na safe yung anak nila. Kadalasan kailangan ko pang hanapin sa logbook at sabihin sa kanila isa-isa kasi walang ibang record na mabilis makita.

Interviewer: Kapag may event o emergency, paano niyo nalalaman kung sino pa ang nasa loob ng school?

Interviewee: Kailangan ko i-check isa-isa yung mga bata para malaman kung sino ang andun at sino ang wala. Medyo matagal minsan kasi maraming bata at maraming service ang sabay-sabay dumating.

Interviewer: Sa tingin niyo, ano ang pinakamalaking dahilan kung bakit minsan matagal ang dismissal process, at paano niyo gustong ma-improve ang sistema?

Interviewee: Minsan, dahil sabay-sabay dumadating ang mga service at manual pa rin ang pag-check ng bawat bata, natatagal ang proseso. Siguro mas maayos kung mas maayos yung coordination sa gate at may paraan para mas malinaw kung sino ang sinundo na at sino pa ang naghihintay.

APPENDIX C.

Curriculum Vitae of

Jade Michael D. Godalle

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godallejade69@gmail.com
09637425619

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	September 2022 - Present	STI College Munoz-EDSA
Vocational/Technical	N/A	
High School	June 2016 - March 2022	Bulan National High School
Elementary	June 2010 - March 2016	Bulan North Central School - B

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
N/A	N/A	N/A

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
Basic Programming	Conscious Competence	Year 2022

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
Year 2025	Beyond Firewalls: Strengthening People and Systems for a Secured Digital Future
Year 2023	Tagisan ng Taleno - Codefest

Curriculum Vitae of

Edson John R. Solitario

34 Union Ext., Barangay Culiat, Quezon City, Metro Manila, Philippines
edsonsolitario246@gmail.com
09951648943

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	September 2022 - Present	STI College Munoz-EDSA
Vocational/Technical	N/A	
High School	June 2013 - March 2022	STI College of Ormoc
Elementary	June 2007 - March 2013	New Era Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
N/A	N/A	N/A

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
Preventive Maintenance	Conscious Competence	Year 2022

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
Year 2025	Beyond Firewalls: Strengthening People and Systems for a Secured Digital Future

Curriculum Vitae of

Francis Jay D. Raagas

370 Lucas Cuadra, Caloocan City, Metro Manila, Philippines
francisraagas44@gmail.com
09614066285

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	September 2022 - Present	STI College Munoz-EDSA
Vocational/Technical	N/A	
High School	June -2016 - March 2022	Ismael Mathay Senior High School
Elementary	June 2010 - March 2016	Sta. Quiteria Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
N/A	N/A	N/A

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
Basic Programming	Conscious Competence	Year 2022
		month year
		month year

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
Year 2025	Beyond Firewalls: Strengthening People and Systems for a Secured Digital Future

Curriculum Vitae of

Laurence Emmanuel M. Supangan

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laurencesupangan@gmail.com
09217871567

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	September 2022 - Present	STI College Munoz-EDSA
Vocational/Technical	N/A	
High School	June 2016 - March 2022	San Francisco High School
Elementary	June 2010 - March 2016	Esteban Abada Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
Year 2021	Social Media Manager	Barangay San Antonio Hall

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

SKILLS

SKILLS	Level of Competency	Date Acquired
Basic Programming	Conscious Competence	Year 2022
Photo Editing	Conscious Competence	Year 2022
Video Editing	Conscious Competence	Year 2022

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
Year 2025	Beyond Firewalls: Strengthening People and Systems for a Secured Digital Future
Year 2025	Catholic Youth Leaders Conference
Year 2024	Catholic Social Teachings